

Chains of reasoning - Set 1 - Solutions

Specification Prf2, Prf3, Prf5.

Answer key

Question	Answer	Topic
1	A	Chains of reasoning
2	B	Chains of reasoning
3	A	Chains of reasoning
4	D	Chains of reasoning
5	F	Chains of reasoning
6	E	Chains of reasoning
7	D	Chains of reasoning
8	D	Chains of reasoning
9	D	Chains of reasoning
10	E	Chains of reasoning
11	C	Chains of reasoning
12	F	Chains of reasoning

Worked solutions

Question 1 RSN-05 · Chains of reasoning · spec Prf5

In a knockout tournament with 37 players, each match eliminates exactly one player and the tournament ends when one player remains. How many matches are played?

Answer: A

Each match eliminates one player and 36 must be eliminated, so 36 matches - whatever the draw looks like.

Question 2 RSN-07 · Chains of reasoning · spec Prf5

The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

Answer: B

The three rows together contain every digit once, so $3 \times (\text{row sum}) = 1 + 2 + \dots + 9 = 45$, giving 15.

Question 3 RSN-10 · Chains of reasoning · spec Prf5

Each of 100 cards has a different integer from 1 to 100 written on it. What is the smallest number of cards that must be drawn (without looking) to be certain of having two cards whose numbers sum to 101?

Answer: A

The 50 pairs $\{1, 100\}, \{2, 99\}, \dots, \{50, 51\}$ each sum to 101. Fifty cards (one from each pair) may avoid it; 51 cannot.

Question 4 RSN-14 · Chains of reasoning · spec Prf5

In a class every student studies at least one of Maths and Physics. 70% study Maths and 60% study Physics. A student is picked at random. What is the probability they study both?

Answer: D

$P(M \text{ or } P) = 1 = 0.7 + 0.6 - P(\text{both})$, so $P(\text{both}) = 0.3$.

Question 5 RSN-16 · Chains of reasoning · spec Prf5

The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?

Answer: F

$(m - 4) + (m - 2) + m + (m + 2) + (m + 4) = 5m$ where m is the middle (odd) number; $5m$ is odd, so 10 is never a factor.

Question 6 RSN-21 · Chains of reasoning · spec Prf5

Seven consecutive integers are written down. Which of the following statements is/are true?

- I Their sum is divisible by 7.
- II Their product is divisible by 7.
- III Their sum is odd.

Answer: E

The sum is $7m$ where m is the middle integer (I). Seven consecutive integers cover every remainder mod 7, so one is a multiple of 7 (II). $7m$ is odd only when m is odd, so III need not hold.

Question 7 RSN-22 · Chains of reasoning · spec Prf5

A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

Answer: D

Each of the 4 rows contains exactly 2 black squares: 8 in total.

Question 8 RSN-01 · Chains of reasoning · spec Prf5

A selection of n distinct integers is taken from 1, 2, ..., 20. What is the smallest n that guarantees two of the chosen integers differ by exactly 10?

Answer: D

Pair up $\{1, 11\}, \{2, 12\}, \dots, \{10, 20\}$: ten pairs. Choosing 11 numbers forces two from the same pair (pigeonhole); choosing one from each pair gives 10 numbers with no such difference.

Question 9 RSN-03 · Chains of reasoning · spec Prf5

Five friends each shake hands with some of the others (each pair shakes hands at most once). Which of the following is impossible?

Answer: D

If four people each shook hands with all four others, everyone shook hands with everyone, so the fifth person also made 4 handshakes, not 1. The others can happen: B is a pentagon of handshakes, C is everyone with everyone, D is a chain, E leaves one person out.

Question 10 RSN-04 · Chains of reasoning · spec Prf5

x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

Answer: E

$(x - y)(x + y) = 45$ with $0 < x - y < x + y$: $(1, 45), (3, 15), (5, 9)$ give $(x, y) = (23, 22), (9, 6), (7, 2)$. Three pairs.

Question 11 RSN-06 · Chains of reasoning · spec Prf5

Three statements are made about a positive integer N :

- N is a perfect square.
- N is a multiple of 6.
- N is less than 50.

Exactly two of the statements are true. Which of the following could NOT be N ?

Answer: C

36 makes all three statements true, not exactly two. Each of the others makes exactly two true: 16 and 25 (square, < 50), 48 and 12 (multiple of 6, < 50), 144 (square, multiple of 6).

Question 12 RSN-08 · Chains of reasoning · spec Prf5

a , b and c are real numbers with $a < b < c$. Which of the following statements must be true?

I $a + b < 2c$

II $ab < bc$

III $\frac{a+c}{2} < c$

Answer: F

I: $a + b < c + c$. III: $\frac{a+c}{2} < \frac{c+c}{2} = c$. II fails when b is negative: $a = -3$, $b = -2$, $c = -1$ gives $ab = 6 > bc = 2$.